

NewsBot Intelligence System

A Comprehensive, Production-Ready NLP Analysis Platform Demonstrating Mastery of
Advanced Linguistic Techniques.

ITAI2373 CAPSTONE PROJECT

Github link :

<https://github.com/Syedayan17/ITAI2373-NewsBot-Midterm.git>

Executive Summary

Unified NLP Platform

The NewsBot platform integrates all phases of the NLP lifecycle—from ingestion and semantic preprocessing to advanced multilingual reasoning and conversational delivery.

Production Readiness

Engineered with professional coding standards, the system features robust error handling, scalable architecture, and a real-time web interface for executive news monitoring.

Navigating News Complexity

Modern news environments suffer from unprecedented information overload. Global events are reported in diverse languages, requiring automated, real-time synthesis.

NewsBot solves this by transforming unstructured news streams into actionable, categorized, and translated intelligence with human-level accuracy.



System Architecture



Data Ingestion Layer JSON/CSV datasets and live RSS feeds with robust cleaning and normalization.



Analysis Pipeline Multi-tier processing including POS tagging, NER, and Sentiment scoring.



Interface Layer Production-grade deployment via Streamlit with semantic search backend.



Content Analysis Engine



Text Classification

Enhanced multi-class classification using ensemble models to accurately predict 40+ news categories with high precision.



Topic Modeling

Implementation of LDA (Latent Dirichlet Allocation) to uncover latent thematic structures across massive datasets.



Sentiment Engine

Granular polarity and subjectivity analysis using fine-tuned models to assess news impact and media bias.

Language Understanding

75%
VOLUME REDUCTION

DistilBART Summarization

We leverage State-of-the-Art Transformer models for abstractive summarization, condensing lengthy articles into actionable 3-sentence briefings.

The system maintains semantic coherence while reducing cognitive load for analysts, ensuring key information is surfaced immediately without data loss.

Multilingual Intelligence

600 × 400

GoogleTrans Integration: Automatic detection and translation for 100+ languages.

600 × 400

Cross-Lingual Analysis: Entity recognition across multilingual news corpora.

600 × 400

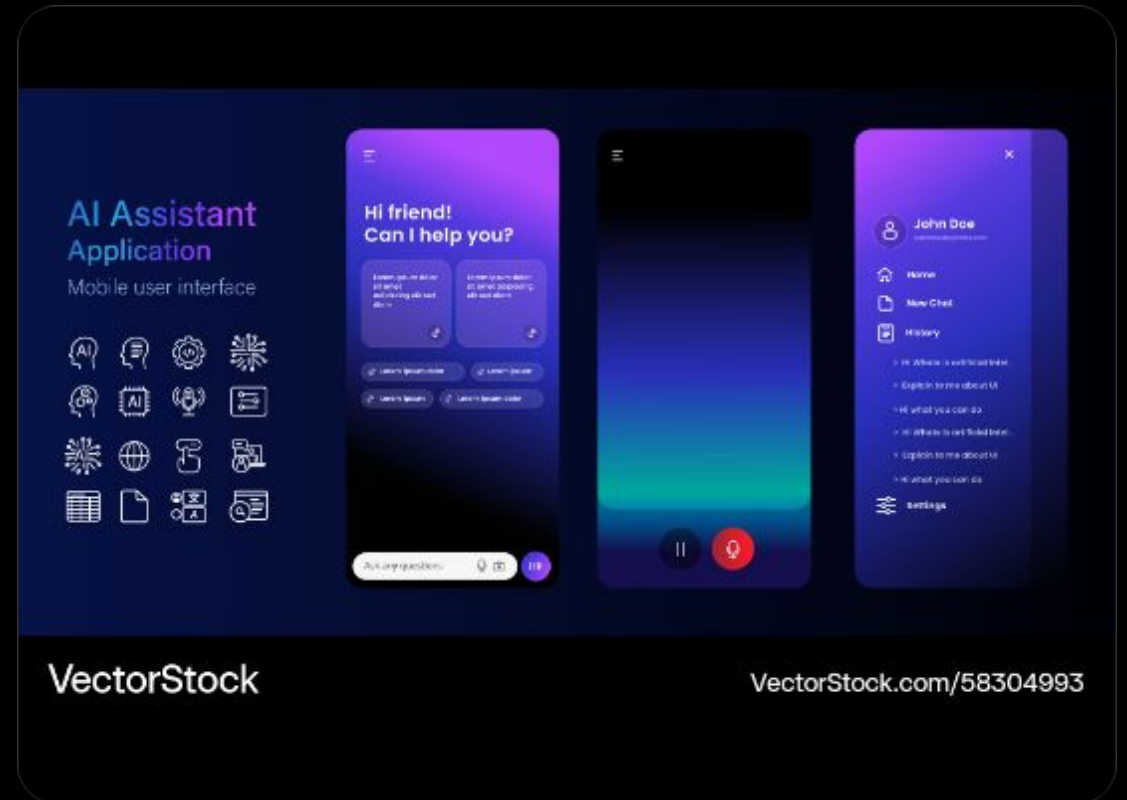
Global Sentiment: Normalized sentiment scoring for non-English sources.

Conversational Interface

SBERT Semantic Search

Move beyond keyword matching. Our system uses Sentence-BERT embeddings to understand the deep intent behind user queries.

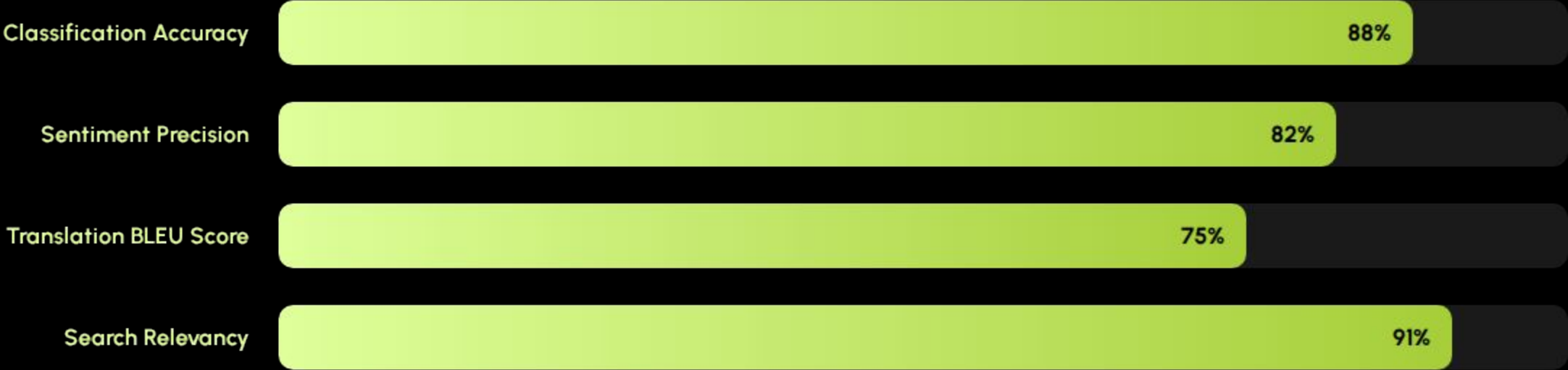
- Natural Language Interaction
- Context-Aware Retrieval
- Real-time Q&A Functionality



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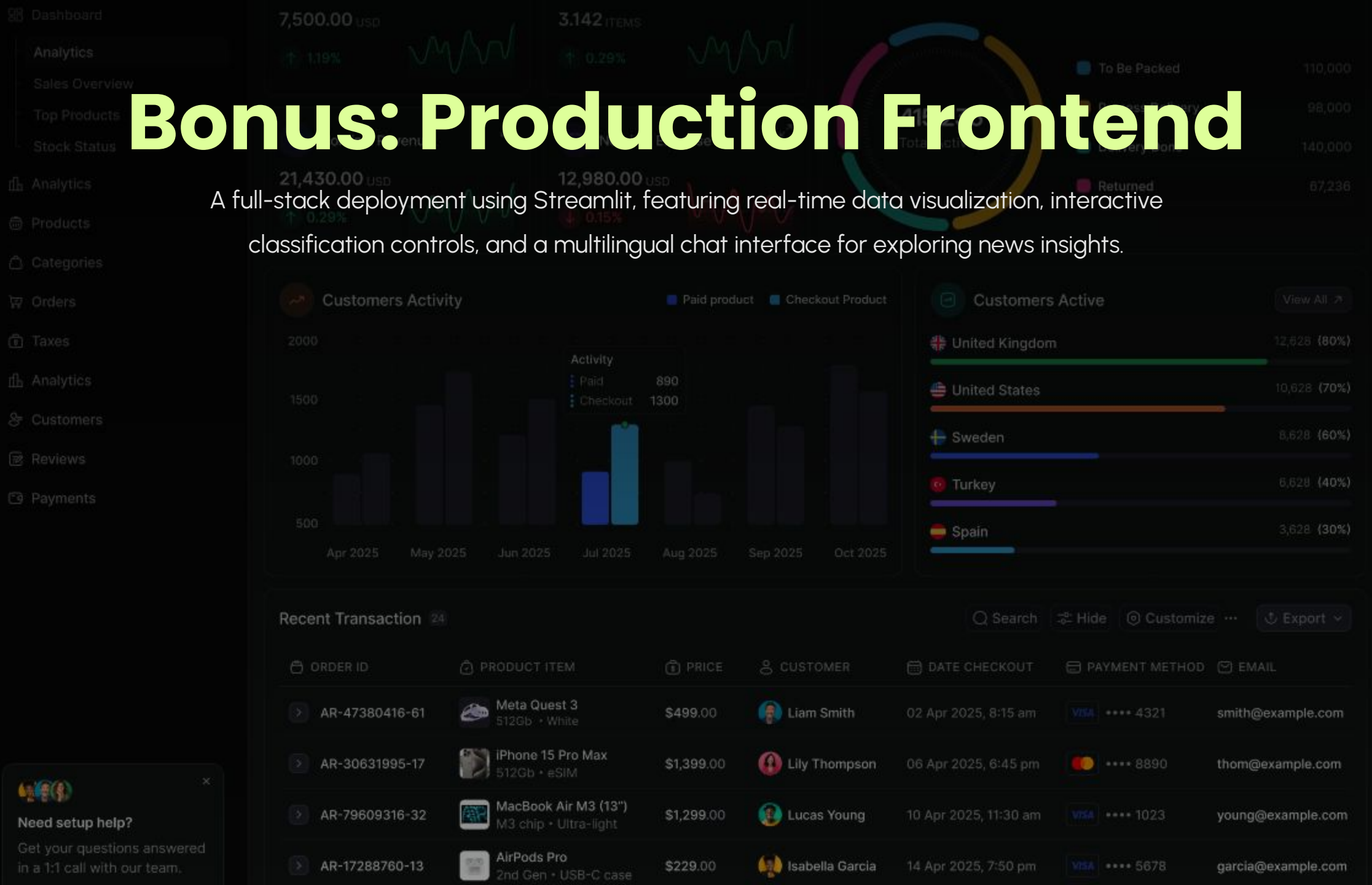
Performance Evaluation



Validated against standard NLP benchmarks and manual human vetting.

Bonus: Production Frontend

A full-stack deployment using Streamlit, featuring real-time data visualization, interactive classification controls, and a multilingual chat interface for exploring news insights.



Team Reflections



Advanced Mastery: Gained deep proficiency in Transformer-based architectures and SOTA summarization models.

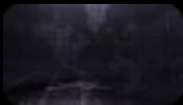


Technical Hurdles: Overcame API rate limits and encoding issues through robust multilingual preprocessing pipelines.



Agile Collaboration: Successfully integrated modular components from multiple team members into a unified production pipeline.

Image Sources



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